VAST Challenge 2021: Mini-Challenge 2

Sean Cheng (sean.cheng@tufts.edu)
Stephanie Kim (stephanie.kim640944@tufts.edu)
Ian Unebasami (ian.unebasami@tufts.edu)

Abstract

Abstract (1-2 paragraphs): The abstract should include your motivation for this project, what you are trying to do, and what you want to achieve.

In this challenge, we are dealing with the mysterious disappearances of several employees from the company GAStech, a natural gas production company from Kronos that has seen monetary success in their business but backlash for their lack of environmental stewardship. It is suspected that the organization, Protectors of Kronos, is responsible for these disappearances, but ultimately, it is our responsibility to find out what truly happened.

Our group chose to do Mini-Challenge 2, which uses geospatial temporal data to identify behavioral patterns and anomalies. We found this challenge interesting because we have never used geospatial temporal data in our class work before, so we thought it would be an interesting challenge to work with new data. We also thought it would be a great chance to pair it with Tableau, a tool we learned to use earlier this semester, so that we are able to plot specific locations at specific times.

Our goal with this project is to develop a visual analytics system that uses geospatial temporal data to track the locations and behavioral patterns of various GStech employees and understand some of their mysterious disappearances. Ultimately, through this system, we want to understand which GAStech employees made which purchases and identify suspicious patterns of behavior by matching the employees to their GPS locations, credit card purchases, and timestamps.

Data Description

Data Description (1-2 paragraphs): Summarize the data. Show that you have performed an initial exploration of the data.

An initial exploration of the dataset reveals three main data sources: vehicle data, transaction records, and geolocation tracking. The car-assignments table includes first and last names, unique Car IDs, and job-related details such as department and title. The GPS dataset logs vehicle movement via timestamps and GPS coordinates (longitude and latitude), enabling temporal and spatial analysis of car locations across time. This can be used to monitor travel patterns, identify frequent destinations, and detect anomalies. But there is no clear relationship between GPSlocations and business names provided in the credit card and loyalty card datasets.

The transaction data is split into loyalty card and credit/debit card records, both containing timestamps, business location names, and purchase amounts. Loyalty card transactions are linked via a unique "L" number, while credit/debit card transactions are linked via the last four digits of the card numbers. We also see overlaps between the loyalty card and credit/debit card transactions because loyalty-card purchases are made with the credit/debit cards. These datasets, when cross-referenced with vehicle movements and employee data, can help trace individual behavior, spending patterns, and location overlaps. Additionally, the inclusion of ESRI shapefiles and tourist maps of Kronos and Abila allows for spatial visualization and contextual analysis, such as identifying if employees visited tourist spots or if vehicles were near points of interest. This comprehensive combination of datasets supports robust geographic and behavioral investigation.

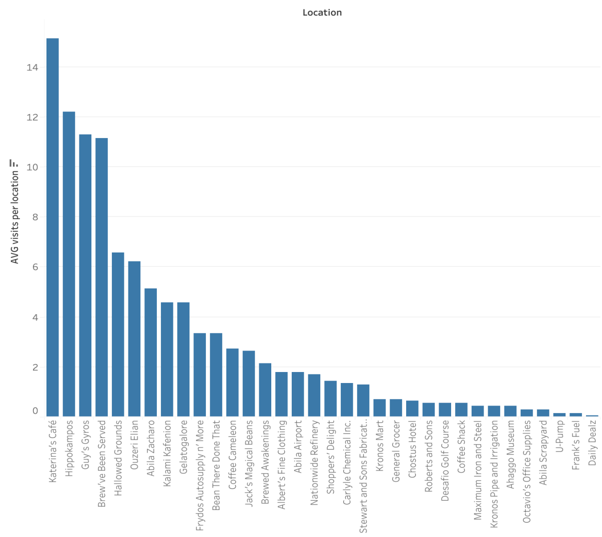
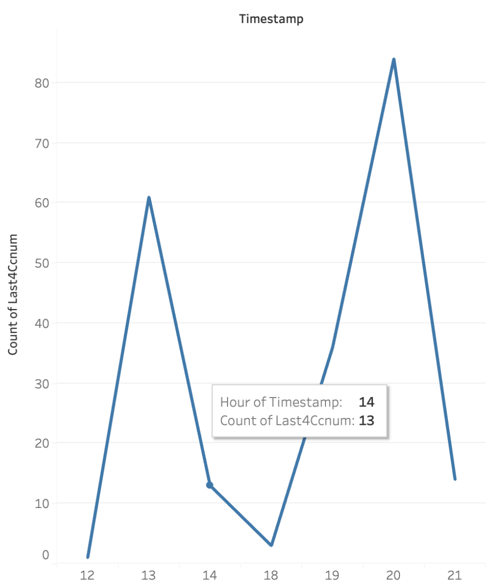
Definition of Goals and Tasks

Definition of Goals and Tasks of your final project (1-2 pages). Show that, in addition to having explored the data, you understand the goals and tasks of your challenge. For each question, first pose (at least) one hypothesis of the ground truth answer, then describe how you plan to prove or disprove it.

1. Using just the credit and loyalty card data, identify the most popular locations, and when they are popular. What anomalies do you see? What corrections would you recommend to correct these anomalies?

   Assumption:
   We are using only the cc_data.csv as the assumption is that a loyalty card is only used if credit cards are used–the two datasets overlap.

   Hypothesis: We think the most popular locations are:
   1. Katerina - most popular at 8PM, least popular at 2pm to 6pm
   2. Hippokampus - most popular at 1PM, least popular at 2pm
   3. Guy’s Gyros - most popular at 8pm, least popular 2pm
   4. Brew’ve Been Served - most popular 7AM, least popular 8AM

	This is how we identified the most popular spots. We did this by averaging the frequency of visits across all 14 days (from Jan 6 to Jan 19)
	This is an example graph of the most popular cafes' visits by hour. This is Katerina's Cafe's graph. We would document that it's typically least visited at 6PM and most visited at 8PM. We could then pick out certain days for Katerina's Cafe where it receives higher amounts of visits at 6PM and lower amounts of visits at 8PM as anomalies. Anomalies will be recorded as a form of "correction".

2. Add the vehicle data to your analysis of the credit and loyalty card data. How does your assessment of the anomalies in question 1 change based on this new data? What discrepancies between vehicle, credit, and loyalty card data do you find?

   Hypothesis: We expect vehicle GPS tracking data should match a person's credit and loyalty card use. This will be somewhat tricky due to not knowing the exact coordinates of each establishment, since the credit card data uses location names and GPS data provides latitude and longitude. The car-assignments data provides a name that we can then match to each loyalty and credit card purchase. We may find anomalies where a vehicle is not at the same location as the matched credit and loyalty card transaction took place.

Our Plan: Our understanding of anomalies has expanded to a person being in a location for a long duration, which is defined as a higher than usual car idle time (where cars show minimal movement in GPS data despite not having any purchases). These patterns could indicate unauthorized meetings or non-business activities. Once matched credit card purchases to GPS tracking data by credit card number and vehicle ID, we can use that information to know who that person is using the vehicle assignments.

3. Can you infer the owners of each credit card and loyalty card? What is your evidence? Where are there uncertainties in your method? Where are there uncertainties in the data? Stephanie to finish

   Hypothesis: We plan to infer the owners of each credit card and loyalty card by matching the timestamps of the card purchases to the timestamps of the GPS data. From there, we can crossmatch the card owners to the names that the cars are assigned to. However, this runs under the assumption that employees are always using the cars assigned to them and making purchases on their own card. Anomalies would include inconsistencies to these assumptions. However, this process has some uncertainty. Firstly, the GPS data tracks the date and time to the second while the credit card data only tracks date and time to the minute and loyalty card data, only by date. The precision gets lost due to lack of consistent timestamp formatting. Secondly, we have the names of some truck drivers who do not have car IDs assigned to them but have their purchases being recorded, so not all credit cards and loyalty cards might be matched.

   Our Plan: As mentioned above, our plan is to match the timestamps of the card purchases to the timestamps of the GPS data (they might not be exact matches, so we would have to account for a buffer) and then match the car IDs from the GPS data to the names in car_assignments.csv. We hope to accomplish this by creating a time aligned map that plots the purchases and GPS locations, where we can watch how the positions shift over specific periods of time.

4. Given the data sources provided, identify potential informal or unofficial relationships among GASTech personnel. Provide evidence for these relationships.

   Hypothesis: We expect to find work friendships between coworkers who frequent the same places outside of working hours such as coffee in the morning or restaurants in the evening. Those meet-ups can be traced using both GPS tracking data and credit card purchase data. There may be potential closer relationships to be discovered between coworkers who appear to spend the night at the same location either due to being roommates or being romantically involved which can only be found by GPS data of two different cars appearing to be parked overnight at the same location.

   Plan: Using GPS and credit card data, we can sequentially find occurrences where clusters of people gather outside of working hours. If the same people end up in the same place at the same time several times, that would provide evidence that those people have some relationship outside of the workplace.

5. Do you see evidence of suspicious activity? Identify 1- 10 locations where you believe the suspicious activity is occurring, and why.

   Hypothesis: We expect to find evidence of suspicious activity by identifying locations where GAStech employees' behavior deviates from normal patterns. Suspicious activities could include vehicles lingering at locations without any corresponding purchases, unusual timing of visits outside normal business hours, or repeated visits to the same suspicious location by the same employee. We hypothesize that by combining the GPS vehicle tracking data with the credit and loyalty card transaction data, we will be able to highlight between one and ten key locations where these suspicious behaviors are taking place. These patterns will help us uncover possible unauthorized meetings, non-business activities, or other irregularities among GAStech personnel.

   Plan:

   We will use Python for data cleaning and preprocessing, and Tableau for visual analytics to explore suspicious activity patterns. First, we will clean and prepare the data using Python. We will read in the gps.csv, cc_data.csv, loyalty_data.csv, and car-assignments.csv files using Pandas. We will parse and clean the timestamps with pd.to_datetime() and extract separate fields for date and hour to support time-based analysis later. After loading the data, we will create mappings by matching vehicle IDs to employee names based on the car-assignments.csv file and link transaction records to specific locations and times.

   Next, we will prepare the GPS points by identifying periods where vehicles are idling, defined as periods where the vehicle speed is near zero for an extended time. We will then match these vehicle stops to known business locations, either through matching names directly or using approximate latitude and longitude coordinates. Once these steps are complete, we will tag potential anomalies by flagging locations where vehicles visit without a corresponding credit or loyalty card purchase and highlighting any timestamps that fall significantly outside of normal business hours (for example, visits before 6 AM or after 9 PM).

   After preprocessing and tagging potential anomalies, we will import the cleaned datasets into Tableau, including the GPS logs, transaction logs, and matched car assignments. In Tableau, we will systematically search for evidence of suspicious activity. Our main method will be to create a time-based activity heat map that shows the density of vehicle visits by hour and location to look for spikes in activity at unusual times.

   Next, we will analyze the visit frequency to unusual or nonstandard locations by identifying places that are not common workplaces or cafés and ranking locations based on high visit frequency but low purchase activity. We will also detect repeated visits by highlighting locations where the same vehicles or employees visit multiple times outside of normal business hours. Finally, we will cluster employees based on GPS data by identifying instances where multiple vehicles are located at the same place at the same time, which could suggest informal gatherings or unofficial meetings.

By combining these techniques, we aim to identify 1 to 10 suspicious locations supported by clear evidence from GPS tracking, transaction patterns, and employee behavior outside of work norms.

Feature List

Feature List. Describe tasks, visualizations, tools, analyses, etc. that you will showcase in your final project. Label them as must-have, good-to have, and optional.

Must-Have Features

1. Bar Charts and Heatmaps: Display location popularity and temporal patterns (addressing Question 1).
   - Example: Show most popular cafés and peak visiting hours.
2. Time-Aligned Map: Map car placements and purchases dynamically based on time (addressing Question 2).
   - Example: Animated map showing car movements and purchase locations minute-by-minute.
3. Suspicious Activity Detection Log: Highlight and log unusual visits, including cases where employees are found at unexpected locations, stay for long periods without purchases, or visit outside typical business hours (addressing Question 5)

Good-to-Have Features

1. Confidence Interval Estimates: Show confidence levels when inferring credit and loyalty card ownership from GPS proximity and timing (addressing Question 3).
2. User and Car ID Filters:
   a. Allow interactive filtering on specific users, car IDs, locations, and time ranges.
3. Time Slider:
   a. Provide a day and time slider to step through events sequentially across the two-week dataset.

Optional Features

1. Clustering Analysis: Detect informal relationships by clustering employees who visit the same location together outside working hours (addressing Question 4).
2. Manual Annotation System: Enable users to manually flag or comment on suspicious behaviors directly within the visual analytics system.

Project Timeline

Project Timeline (with milestones when you are planning to finish which feature)
4/14 — Submit Initial Report (Part 1)
4/14-17 — Plan out visual design and layout:

- Decide what types of information to display (e.g., visits per location, time activity patterns, suspicious activity).
- Sketch draft versions of key visualizations (barcharts, heatmaps, time-aligned maps).

4/18-25 — Implement visualizations in Tableau:
- Build barcharts, heatmaps, and interactive time-aligned maps.
- Add user and car ID sliders, and location filters to support data exploration.

4/26-5/2 — Analyze data for suspicious activity:
- Identify and document potential suspicious locations.
- Form a rough draft of the final project report.

5/3-4 — Finalize report
- Clean up all visualizations.
- Refine analysis write-ups.
- Double-check all findings and prepare presentation/demo materials.

5/4 — Submit final project.
5/5 — Live demo

Team Roles

Team Roles: who's going to be doing what in your team?

All team members will work together to:
1. Finalize and polish the visual dashboards in Tableau.
2. Help review and edit the final project report.
3. Prepare talking points and practice for the live demo presentation.

We will collaborate using shared Google Docs for writing, Python Notebooks for cleaning and preprocessing, and Tableau Public for visual dashboard sharing. Weekly check-ins will be used to track progress and reassign tasks if needed.

Ian: Data Cleaning, Transaction and GPS Preprocessing, and Popularity Visualization
Responsibilities:
- Import and clean all datasets (cc_data.csv, loyalty_data.csv, gps.csv, car-assignments.csv) using Python.
- Parse timestamps, extract dates and hours, and merge datasets where needed.
- Preprocess GPS data to clean location points, normalize timestamps, and prepare vehicle datasets.
- Analyze credit card and loyalty card transactions to identify the most popular locations and peak visiting hours.
- Create bar charts and heatmaps in Tableau to visualize location popularity and temporal patterns.
- Identify and document initial anomalies (unexpected activity patterns).
- Import and clean all datasets (cc_data.csv, loyalty_data.csv, gps.csv, car-assignments.csv) using Python.

- Parse timestamps, extract dates and hours, and merge datasets where needed.
- Analyze credit card and loyalty card transactions to identify the most popular locations and peak hours.
- Create bar charts and heatmaps in Tableau showing location popularity and temporal patterns.
- Identify and document initial anomalies (unexpected visit times or transaction spikes).

Related Questions:

- Question 1: Popular locations, time patterns, initial anomalies.

Sean: Vehicle-Purchase Matching, Suspicious Activity Detection, and Time-Aligned Map Creation

Responsibilities:

- Analyze GPS tracking data to detect long vehicle idle periods (minimal movement for extended time).
- Cross-match vehicle GPS locations with purchase transactions (credit and loyalty).
- Detect suspicious behavior (vehicle at a location with no purchase, vehicles at unusual times, or frequenting odd locations).
- Build the time-aligned map in Tableau to show vehicle movement and purchases.
- Create the suspicious activity detection log that identifies 1–10 key suspicious locations.

Related Questions:

- Question 2: Assessing anomalies with vehicle data.
- Question 5: Identifying suspicious activities and locations.

Stephanie: Card Ownership Inference, Relationship Clustering Analysis, and Final Report Writing

Responsibilities:

- Infer the owners of each credit card and loyalty card based on GPS proximity and transaction matching.
- Investigate potential informal relationships by detecting employee clustering at non-work locations after hours.
- Support building clustering analysis dashboards (optional feature) if time permits.
- Lead the writing and organization of the final project report and preparation for the live demo presentation.

Related Questions:

- Question 3: Inferring card and loyalty card owners.
- Question 4: Identifying informal relationships among employees.